

Quantum Teleportation: Protocol, Implementation, and Applications

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Abstract

This report provides a comprehensive study of quantum teleportation, covering its protocol, implementation, and applications. We begin with an introduction to the fundamental concepts and historical development of quantum teleportation. We then present the theoretical framework and the step-by-step protocol for achieving quantum teleportation. Following this, we discuss various implementation strategies and their practical challenges. Finally, we explore the current and potential applications of quantum teleportation in quantum information processing and communication.

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I. INTRODUCTION

II. MATHEMATICAL BACKGROUND

A. Qubits and the Hilbert Space Formalism

The fundamental unit of quantum information is the "qubit". Unlike classical bit, which takes the value of either 0 or 1 at one of the time, a qubit exists in a two-dimensional Hilbert space $\mathcal{H} = \text{span}\{|0\rangle, |1\rangle\} \cong \mathbb{C}^2$, and its state can be described by a unit vector with an orthonormal computational basis $\{|0\rangle, |1\rangle\}$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1)$$

where α and β are the probability amplitudes: measurement yielding $|0\rangle$ or $|1\rangle$ has a probability $|\alpha|^2$ and $|\beta|^2$, respectively [1]. Unlike classical bit, a qubit can exist in a superposition of both states, which has no classical analogue. Physical realization of qubits include the two polarization of photon and two energy levels of an electron in an atom.

Bloch Sphere As an phase factor $e^{i\gamma}$ does not have observable effect, Eq. 1 can be written in a more general expression

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi) \quad (2)$$

The two parameters θ and φ define a unique point on the unit three-dimensional sphere, known as the Bloch sphere as shown in Figure 1. The north pole ($\theta = 0$) corresponds to $|0\rangle$, while the south pole corresponds to $|1\rangle$, and every point on the equator represents an equal superposition such as $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Multi-qubit systems. For a system of two qubits, we use the tensor product of the individual Hilbert space $\mathcal{H}_{12} = \mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathbb{C}^4$ to represent the state space, with computational basis kets $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Here we use the ordering convention. That is, $|01\rangle = |0\rangle_1 \otimes |1\rangle_1$. Similarly, if it is a system of three qubits, then $|010\rangle = |0\rangle_1 \otimes |1\rangle_2 \otimes |0\rangle_3$. A general two-qubit state takes the form

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle, \quad \sum_x |\alpha_x|^2 = 1 \quad (3)$$

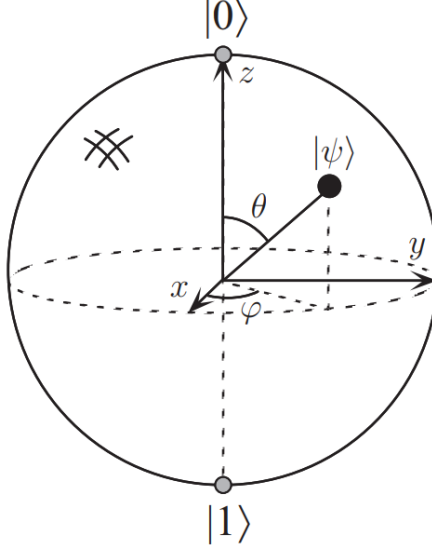


FIG. 1: Bloch sphere schematic. An arbitrary qubit can be represented by one point on the sphere [1].

B. Entanglement and the Bell States

A two-qubit state is called separable if it can be factored as $|\psi\phi\rangle = |\psi\rangle_A \otimes |\phi\rangle_B$. This means that qubit A is in state $|\psi\rangle$, while qubit B is in state $|\phi\rangle$. On the other hand, an entangled state can not be factorized into the product of two single-qubit states. In this case, the quantum state can only be understood as a property of the whole system. The most important entangled states for quantum teleportation are the Bell states [1, 2]

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), & |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \end{aligned} \tag{4}$$

These four states form an orthonormal basis for the two-qubit Hilbert space. Besides, they are maximally entangled: tracing out either qubit leaves the reduced density matrix $\rho = I/2$, where I is the identity matrix. For example, consider the Bell state $|\Phi^+\rangle_{AB}$, then the corresponding density matrix is $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$. Now, if we only look at the qubit A subsystem, we trace out qubit B to get the reduced density matrix as

$$\rho_A = \text{tr}_B(\rho_{AB}) = \frac{I_A}{2}. \tag{5}$$

Similarly, we have $\rho_B = \frac{I_B}{2}$. This implies that when measuring either qubit, we get equal probability of getting $|0\rangle$ or $|1\rangle$. However, the measurement results of system A and system B are perfectly correlated. For example, if system A measure the entangled state and gets $|0\rangle$, then B gets $|0\rangle$. Similarly, if A obtains $|1\rangle$, then B gets $|1\rangle$. That is, each local measurement result is random, but it becomes highly correlated once we look at the system as a whole.

C. Quantum Measurement and Bell Measurement

The projection postulate state that measurement irreversibly collapses the superposition state into one of its components [3]. Therefore, measurement on the qubit is a physical process that will disturb the original state. A projective measurement is described by a Hermitian observable $M = \sum_m m P_m$, where $\{P_m\}$ are orthogonal projection operator onto the eigenspace of M with eigenvalue m and satisfy $\sum_m P_m = I$ and $P_m P_{m'} = \delta_{mm'} P_m$. Given a pure state $|\psi\rangle$, the probability of obtaining outcome m is

$$p(m) = \langle \psi | P_m | \psi \rangle \quad (6)$$

and the post-measurement state collapses to [1]

$$|\psi_m\rangle = \frac{P_m |\psi\rangle}{\sqrt{p(m)}}. \quad (7)$$

For example, consider a qubit $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. If we do the measurement in the computational basis $\{|0\rangle, |1\rangle\}$, then the corresponding projection operators are $P_0 = |0\rangle \langle 0|$, $P_1 = |1\rangle \langle 1|$. The probability and the post-measurement states of the two possible outcomes are

$$\begin{aligned} p(0) &= \langle \psi | P_0 | \psi \rangle = |\alpha|^2, & P_0 |\psi\rangle &= |0\rangle \langle 0| (\alpha |0\rangle + \beta |1\rangle) = \alpha |0\rangle \\ p(1) &= \langle \psi | P_1 | \psi \rangle = |\beta|^2, & P_1 |\psi\rangle &= |1\rangle \langle 1| (\alpha |0\rangle + \beta |1\rangle) = \beta |1\rangle. \end{aligned} \quad (8)$$

Therefore, a direct measurement gives either 0 or 1, and the original coefficients disappear due to the normalization (Equation 7), and thus the original qubit state is no longer available. In order to maintain the original qubit state, in quantum teleportation, a Bell measurement is performed on a two-qubit system. Similar to the projective measurement, we define the

Bell-measurement projectors as

$$P_k = |\beta_k\rangle \langle \beta_k|, \quad |\beta_k\rangle \in \{|\Psi^\pm\rangle, |\Phi^\pm\rangle\}. \quad (9)$$

Thus, a Bell measurement yields one of the Bell states. That is, computational-basis measurement tells us whether each qubit is individually in $|0\rangle$ or $|1\rangle$, while a Bell measurement implies which Bell state the two-qubit system occupies as a whole.

D. Pauli Operators and Unitary Correction

The Pauli operators are a set of 2×2 unitary matrices forming a basis for the space of single-qubit operators [1]:

$$\begin{aligned} I &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & X &= \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\ Z &= \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, & Y &= \sigma_2 = iXZ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \end{aligned} \quad (10)$$

These are the standard single-qubit Pauli operators used in the quantum information. In particular, X is a bit-flip operator ($X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$), Z acts as a phase-flip operator ($Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$), while Y is the same operator as XZ up to a phase factor. Therefore, we will just use XZ in the later protocol operation. These four operators $\{I, X, Z, XZ\}$ will serve as the unitary correction operators for Bob to recover the original quantum state after Alice's Bell measurement.

III. STANDARD QUANTUM TELEPORTATION PROTOCOL

In this chapter, we use the mathematical tools introduced in the previous chapter to derive the standard quantum teleportation protocol. The goal of the protocol is to transfer an unknown quantum state from Alice to Bob without physically sending the original qubit through a quantum channel. Instead, the protocol uses one shared Bell pair, one Bell measurement, two bits of classical communication, and a conditional Pauli correction.

Suppose Alice has an unknown qubit state

$$|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (11)$$

Here the subscript 1 labels the unknown input qubit. Alice does not know the coefficients, and she may not measure it directly as the post-measurement state would collapse. Therefore, Alice and Bob share an entangled Bell pair, say

$$|\Phi^+\rangle_{23} = \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23}), \quad (12)$$

where qubit 2 and 3 are held by Alice and Bob respectively. The total initial state of the three-qubit system is

$$\begin{aligned} |\Psi\rangle_{123} &= |\psi\rangle_1 |\Phi^+\rangle_{23} \\ &= \frac{1}{\sqrt{2}} (\alpha|000\rangle_{123} + \alpha|011\rangle_{123} + \beta|100\rangle_{123} + \beta|111\rangle_{123}), \end{aligned} \quad (13)$$

where in the second line of the equation we expand the total qubit state. Since Alice will perform a Bell measurement on qubit 1 and 2, this suggests us to express the computational basis $|\rangle_{12}$ in terms of the Bell states as

$$\begin{aligned} |00\rangle_{12} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{12} + |\Phi^-\rangle_{12}), & |11\rangle_{12} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{12} - |\Phi^-\rangle_{12}), \\ |01\rangle_{12} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{12} + |\Psi^-\rangle_{12}), & |10\rangle_{12} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{12} - |\Psi^-\rangle_{12}). \end{aligned} \quad (14)$$

Thus, the total state becomes

$$|\psi\rangle_1 |\Phi^+\rangle_{23} = \frac{1}{2} \left[|\Phi^+\rangle_{12} (\alpha |0\rangle_3 + \beta |1\rangle_3) + |\Phi^-\rangle_{12} (\alpha |0\rangle_3 - \beta |1\rangle_3) \right. \\ \left. + |\Psi^+\rangle_{12} (\beta |0\rangle_3 + \alpha |1\rangle_3) + |\Psi^-\rangle_{12} (-\beta |0\rangle_3 + \alpha |1\rangle_3) \right]. \quad (15)$$

We may further rewrite this expression using the Pauli operators as

$$|\psi\rangle_1 |\Phi^+\rangle_{23} = \frac{1}{2} \left[|\Phi^+\rangle_{12} |\psi\rangle_3 + |\Phi^-\rangle_{12} Z |\psi\rangle_3 + |\Psi^+\rangle_{12} X |\psi\rangle_3 + |\Psi^-\rangle_{12} XZ |\psi\rangle_3 \right]. \quad (16)$$

This equation is the central identity of quantum teleportation. It shows that when qubits 1 and 2 are written in the Bell basis, Bob's qubit 3 is related to the original input state by one of four possible Pauli operations. Alice now perform a Bell measurement on qubits 1 and 2. According to the projection postulate, the total state collapses into one of the four Bell states with equal probability. If Alice obtains the outcome $|\Phi^+\rangle_{12}$, then Bob's qubit is simply $|\psi\rangle_3$. With no extra correction, the unknown qubit is teleported to Bob. If Alice get $|\Phi^-\rangle_{12}$, then Bob's qubit is $Z |\psi\rangle_3$. Bob need to apply Z so that $Z(Z |\psi\rangle_3) = Z^2 |\psi\rangle_3 = |\psi\rangle_3$. Similarly, if Alice obtains $|\Psi^+\rangle_{12}$, Bob gets $X |\psi\rangle_3$ and need to perform X so that $X(X |\psi\rangle_3) = X^2 |\psi\rangle_3 = |\psi\rangle_3$. Finally, if Alice obtains $|\Psi^-\rangle_{12}$, Bob apply ZX such that $ZX(XZ |\psi\rangle_3) = |\psi\rangle_3$. These four cases can be summarized into the following table as

Alice's Bell outcome	Bob's state before correction	Bob's correction
$ \Phi^+\rangle_{12}$	$ \psi\rangle_3$	I
$ \Phi^-\rangle_{12}$	$Z \psi\rangle_3$	Z
$ \Psi^+\rangle_{12}$	$X \psi\rangle_3$	X
$ \Psi^-\rangle_{12}$	$XZ \psi\rangle_3$	ZX

(17)

After Alice's measurement, she need to tell Bob which one of these four Bell states she got. Since one classical bit has two possible values, 0 and 1, two classical bits can represent four possible messages: 00, 01, 10, and 11. This is enough to label the four Bell-measurement outcomes. For example, they may agree that 00 corresponds to $|\Phi^+\rangle_{12}$. That is, with a two-classical-bit channel, Alice can transport her measurement result to Bob so that he can perform the corresponding correction operation to get the original unknown qubit state. In

source language, the teleportation protocol can be summarized as

$$1 \text{ ebit} + 2 \text{ classical bits} \rightarrow 1 \text{ transmitted qubit.} \quad (18)$$

IV. PHYSICAL MEANING AND EXPERIMENTAL RELEVANCE

After deriving the standard teleportation protocol, it is important to clarify what the protocol does and does not imply physically. Quantum teleportation transfers an unknown quantum state from Alice's qubit to Bob's qubit, but it does not transfer matter, does not create a second copy of the state, and does not allow faster-than-light communication. The protocol works only because of the shared entangled pair and a classical bit channel. In this section, we discuss these physical implications and then connect the ideal protocol to experimental realization.

A. No-Cloning

Quantum teleportation does not create a second copy of the unknown quantum state. Although Bob's final qubit becomes $|\psi\rangle_3 = \alpha|0\rangle_3 + \beta|1\rangle_3$, Alice's original qubit is no longer available after the Bell measurement. The Bell measurement projects qubits 1 and 2 into one of the four Bell states, and therefore irreversibly destroys the original unknown state on Alice's side. This is why quantum teleportation is consistent with the no-cloning theorem. The no-cloning theorem, first formulated by Wootters and Zurek [4] and independently by Dieks [5], states that it is impossible to make an exact copy of an arbitrary unknown quantum state. The proof is straightforward. Suppose there exists a universal unitary cloning machine U such that $U|\psi\rangle|s\rangle = |\psi\rangle|\psi\rangle$ for any unknown state $|\psi\rangle$, where $|s\rangle$ is a blank initial state. If the same machine can also clone another state $|\phi\rangle$, then $U|\phi\rangle|s\rangle = |\phi\rangle|\phi\rangle$. Taking the inner product of these two equations gives $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$. This condition is only satisfied when $\langle\psi|\phi\rangle = 0$ or $\langle\psi|\phi\rangle = 1$. In other words, the two states must be either orthogonal or identical. Therefore, no single unitary operation can clone an arbitrary unknown quantum state [1].

Teleportation avoids this contradiction because it does not copy the state. Instead, the quantum information initially contained in qubit 1 is transferred to qubit 3, while the original state at Alice's side is destroyed by measurement. Therefore, teleportation should be understood as state transfer rather than state duplication.

B. No Faster-Than-Light Communication

Quantum teleportation also does not allow faster-than-light communication. Although the shared Bell pair provides nonclassical correlations between Alice and Bob, Bob cannot recover the unknown state before receiving Alice's classical message. That is, Bob needs Alice's message so that he knows what operation he should apply to recover the unknown qubit. This can be made more precise by looking at Bob's reduced density matrix before the classical message arrives. Since the four Bell-measurement outcomes occur with equal probability, Bob's local state is an average over the four possible states:

$$\rho_3 = \frac{1}{4} (|\psi\rangle\langle\psi| + Z|\psi\rangle\langle\psi|Z + X|\psi\rangle\langle\psi|X + XZ|\psi\rangle\langle\psi|ZX). \quad (19)$$

For an arbitrary qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, this average becomes

$$\rho_3 = \frac{I}{2}. \quad (20)$$

Thus, before receiving Alice's classical message, Bob's local state is completely mixed and independent of α and β . Any measurement Bob performs locally gives random results and reveals no information about the unknown input state. Bennett et al. [2] emphasized this point in the original teleportation paper: if Bob tries to complete the protocol by guessing Alice's classical message before it arrives, his state is only a maximally mixed state and gives no information about the input state.

C. Security of the Classical Message

V. EXPERIMENTAL IMPLEMENTATION VIA DUAL-PHOTON ENTANGLEMENT

A. The Quantum Leap from Theory to Lab

The 1998 experiment by Boschi et al. marked a paradigm shift in realizing quantum information protocols. While the canonical Bennett et al. (1993) model required three independent photons—posing formidable temporal synchronization hurdles—this design utilized a sophisticated two-photon, dual-degree-of-freedom architecture. By exploiting both the momentum (path or k -vector) and polarization of a single entangled photon pair, researchers accessed a four-dimensional Hilbert space ($2 \text{ paths} \times 2 \text{ polarizations}$) within a bipartite system. This architectural innovation addressed two primary limitations of prior configurations:

- **100% Success Rate in Bell State Measurement:** Unlike linear-optics three-photon schemes (bounded by a 50% success rate), this approach permits projection onto the complete Bell basis, enabling the deterministic identification of all four states simultaneously.
- **Elimination of Three-Photon Coincidence Hurdles:** Deriving the entire process from a single entangled photon pair (EPR source) circumvents the prohibitive difficulty of synchronizing independent sources within sub-nanosecond coincidence windows.

The physical execution initiates at the EPR source, where the non-local quantum channel is established.

B. Implementation of the Dual-Degree-of-Freedom Setup

1. *The EPR Source: Converting Light into Entanglement*

Entanglement is generated via Type-II spontaneous parametric down-conversion (SPDC). Specifically, a 200 mW, 351.1 nm ultraviolet argon laser pumps a β -barium borate (BBO) crystal, producing polarization-entangled photon pairs at 702.2 nm. To facilitate the dual-degree architecture, this initial polarization entanglement is deterministically converted into

spatial path entanglement via birefringent calcite crystals (C), which split the incident photons into discrete spatial paths (a and b). The resulting EPR state is described by:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|a_2\rangle + |b_1\rangle|b_2\rangle)$$

Crucially, photon 1 is routed toward Alice's station, while photon 2 traverses backreflected paths a_2 and b_2 through the BBO crystal toward Bob.

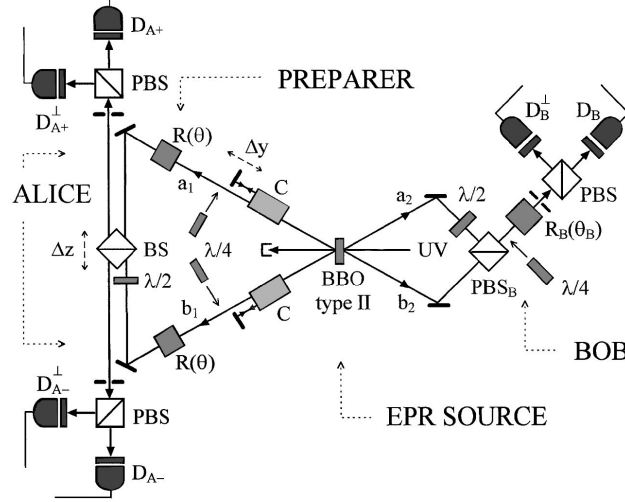


FIG. 2: Diagram of experimental setup showing the separate roles of the preparer, Alice, and Bob.

2. The Preparer: Encoding the Unknown State

A strict logical distinction is maintained between the Preparer and Alice. To ensure the state $|\phi\rangle$ remains fundamentally "unknown," the Preparer encodes it onto photon 1's polarization before it reaches Alice's apparatus. This relies on local unitary operations on paths a_1 and b_1 :

- **Polarization Initialization:** Photon 1 enters with an initial defined polarization (typically vertical, $|v\rangle_1$).
- **Unitary Manipulation:** Quarter-wave plates and Fresnel rhomb rotators (R) transform this into a general pure state: $|\phi\rangle = \alpha|v\rangle_1 + \beta|h\rangle_1$.
- **Path Independence:** Operating identically on paths a_1 and b_1 , the polarization state effectively factors out of the path entanglement.

This yields a global tripartite system where the unknown polarization is carried by the path-entangled photon 1:

$$|\Psi_{total}\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|a_2\rangle + |b_1\rangle|b_2\rangle)(\alpha|v\rangle_1 + \beta|h\rangle_1)|h\rangle_2$$

3. *Alice's BSM Engine: Interferometric Bell State Measurement*

Alice's Bell State Measurement (BSM) interferometer projects photon 1's bipartite state onto four orthogonal path-polarization Bell states:

$$|c^\pm\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|v\rangle_1 \pm |b_1\rangle|h\rangle_1)$$

$$|d^\pm\rangle = \frac{1}{\sqrt{2}}(|a_1\rangle|h\rangle_1 \pm |b_1\rangle|v\rangle_1)$$

Prior to entering a 50:50 beam splitter (BS), a Fresnel rhomb in path b_1 induces a 90° polarization rotation. This ensures path and polarization degrees of freedom are properly aligned for interference. The BS then acts as a unitary operator, mapping distinct path-polarization correlations to specific output ports.

The BS is mounted on a micrometrical stage to precisely tune the path-length difference (Δz). Alignment is confirmed by "interference dips" in coincidence count rates, signifying spatiotemporal wave packet overlap.

Mechanics of Detection: Polarizing beam splitters (PBS) sort the emerging photons into four avalanche photodiode detectors ($D_{A+}, D_{A+}^\perp, D_{A-}, D_{A-}^\perp$). Detector coincidences unambiguously project onto the Bell basis:

- A detection at $D_{A+}^\perp$ or $D_{A-}^\perp$ identifies states $|c^\pm\rangle$.
- A detection at D_{A+} or D_{A-} identifies states $|d^\pm\rangle$.

This projective measurement dictates the unitary operation Bob must perform to reconstruct the initial state.

4. *Bob's Station: Reconstructing the Teleported State*

At the receiving station, photon 2 arrives via trajectories a_2 and b_2 . To resolve the path entanglement, polarization in path a_2 is orthogonally rotated by 90° via a half-wave plate before recombining with path b_2 at the polarizing beam splitter (PBS_B).

Bob employs *passive reconstruction*. Rather than dynamically rotating the state using active electro-optic switches, he reorients his measurement basis to verify the teleported state based on Alice's classical transmission.

Unitary Transformation Logic:

- **Linear States:** Bob adjusts his Fresnel rhomb (R_B) to an angle θ_B , determined by Alice's Bell state outcome.
- **Elliptical States:** Bob inserts a quarter-wave plate at angle γ_B to map the expected elliptical state back into an observable linear basis.
- **Phase Matching:** Bob uses a micrometrical stage to tune Δy , ensuring paths satisfy the phase-matching criteria for high-fidelity verification.

C. The Verdict: Beating the Classical Limit

To rigorously validate quantum teleportation, the experimental fidelity (S)—representing the average probability of the teleported state passing verification—must strictly exceed the classical communication limit of 0.75. Surpassing this threshold empirically proves that entanglement was utilized as a non-local resource. In the 1998 experiment, S was derived through a comprehensive sampling process across three equally spaced linear polarization settings (0° , 120° , and -120°). To ensure a representative result on the Bloch sphere, each of these three states was assigned a weight of $1/3$, while each of the four possible Bell state measurement outcomes was weighted equally at $1/4$. The final observed fidelity of 0.853 ± 0.012 exceeds the classical bound by eight standard deviations. This result robustly confirms that the dual-degree-of-freedom architecture functioned as a unified quantum processor, providing definitive empirical proof of the successful teleportation of unknown pure quantum states.

VI. APPLICATIONS AND REAL-WORLD RELEVANCE OF QUANTUM TELEPORTATION

A. Basic Structure of Report

Prepare a report *in* T_EX and *in* English that contains basically the following sections:

- introduction or motivations of the issue (e.g., what is the problem or question, why this topic is physically interesting, historical notes, etc.);
- model or framework for later discussions or analysis (e.g., defining notations, reviewing basic theory background, providing key equations, etc.);
- derivation in sufficient detail, leaving lengthy stuff to appendices, and discussions of physics in your results (e.g., deriving from the above-mentioned basics to the final results that will answer the initial question, providing more physical understanding about your results, discussing implications, comparing them with empirical data, etc.);
- summary and conclusions (e.g., reminding readers of the highlights in your study/report);
- using figures and tables whenever appropriate or necessary (in particular, you should make it clear whether the plots or numbers are computed by yourselves or copied from the references); and
- bibliography to cite essential and relevant references (learn from professional papers about how to do it properly).

B. About T_EX

C. Writing Style

You should write *in your own words* and provide *critical thoughts* on the topic.

Here are some additional tips about your writing:

- Make good use of modern technology to help you *check spelling and grammatical errors*. Pay attention to the proper usage of punctuation (including those in the equations).

- Be sure to clearly *define your symbols and notations*.
- Be *narrative* rather than simply showing a pile of equations. Provide a physical understanding of factors or terms in the derived results.
- Show sufficiently detailed steps in derivations, so that people at your level of understanding can follow without difficulty. In the case of tedious or trivial algebraic manipulations, you can schematically outline or describe how they are done.
- Role models would be most of the good textbooks and the research papers you have studied.

Appendix A: Useful Formulas

In the appendices, you can provide tedious derivations or detailed discussions that are not crucial for the discussions in the main text.

First, we have for $a > 0$

$$\begin{aligned}
G_0 &\equiv \int_{-\infty}^{+\infty} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \\
G_2 &\equiv \int_{-\infty}^{+\infty} x^2 e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \frac{1}{a} \\
\text{or } G_{2n} &\equiv \int_{-\infty}^{+\infty} x^{2n} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}} \frac{1}{a^n} (2n-1)(2n-3)\cdots 3 \cdot 1
\end{aligned} \tag{A1}$$

in general for $n \in \mathbb{N} \cup \{0\}$, where the factor of $1/2$ in the exponent is common in physics. The proof goes as follows:

$$\begin{aligned}
(G_0)^2 &= \left(\int_{-\infty}^{+\infty} e^{-\frac{1}{2}ax^2} dx \right) \cdot \left(\int_{-\infty}^{+\infty} e^{-\frac{1}{2}ay^2} dy \right) = \int_0^{2\pi} d\phi \int_0^\infty dr r e^{-\frac{1}{2}ar^2} \\
&= 2\pi \cdot \frac{1}{a} \int_0^\infty du e^{-u} = \frac{2\pi}{a} [-e^{-u}]_0^\infty = \frac{2\pi}{a} \\
\Rightarrow G_0 &= \sqrt{\frac{2\pi}{a}} \quad (\text{obviously not the other solution}) .
\end{aligned} \tag{A2}$$

The rest can be proved by induction. Suppose the last formula in Eq. (A1) is true for $n = k$,

then for $n = k + 1$ we have

$$\begin{aligned}
\int_{-\infty}^{+\infty} x^{2(k+1)} e^{-\frac{1}{2}ax^2} dx &= \int_{-\infty}^{+\infty} x^{2k} x^2 e^{-\frac{1}{2}ax^2} dx = \int_{-\infty}^{+\infty} x^{2k} \left(-2 \frac{d}{da}\right) e^{-\frac{1}{2}ax^2} dx \\
&= \left(-2 \frac{d}{da}\right) G_{2k} = \sqrt{2\pi} (2k-1)(2k-3) \cdots 3 \cdot 1 \left(-2 \frac{d}{da}\right) a^{-(k+1/2)} \\
&= \sqrt{2\pi} (2k+1)(2k-1) \cdots 3 \cdot 1 a^{-[(k+1)+1/2]} .
\end{aligned} \tag{A3}$$

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